

1 Definitions

For a function $f : [n] \rightarrow \mathbb{R}$ we say a pair $(i, i + 1)$ for i in $[n - 1]$ is

1. a "good pair" if there exists an optimal LIS that does not change the values in this pair
2. a "bad pair" if no optimal LIS includes this pair untouched

note that all $(j, j + 1)$ such that $f(j) > f(j + 1)$ are "bad pairs"

2 $1 + \delta$ assuming $\varepsilon_f < 0.5$

Theorem 2.1. *If the true ε_f is less than 0.5, Using distance-separation(f, ε, δ), one can check if a function is $\varepsilon_f > \varepsilon$ or $\varepsilon_f < (1/2 - \delta)\varepsilon$ for any epsilon and delta.*